Craniosynostosis and Dental Anomalies (CRSDA / Kreiborg-Pakistani Syndrome): A Comprehensive Disease Characterization

Disease: Craniosynostosis and Dental Anomalies (CRSDA) **MONDO ID:** MONDO:0013615 · **OMIM:** 614188 · **Causal gene: IL11RA** (interleukin 11 receptor subunit alpha, 9p13.3) **Synonyms:** Kreiborg-Pakistani syndrome; IL11RA-related Crouzon-like autosomal recessive craniosynostosis **Category:** Mendelian (autosomal recessive)

Summary

Craniosynostosis and dental anomalies (CRSDA), also known as Kreiborg–Pakistani syndrome, is a rare **autosomal recessive** craniofacial disorder caused by **biallelic loss-of-function variants in IL11RA**, the gene encoding the alpha subunit of the interleukin-11 receptor, located on chromosome **9p13.3**. Despite the "Crouzon-like" clinical overlap with the dominant FGFR-driven craniosynostoses, CRSDA is mechanistically distinct: it is a disorder of **cytokine receptor signaling**, not fibroblast growth factor signaling. The causal gene named in this report — and the gene that must anchor any knowledge-base entry — is **IL11RA**, not an FGFR.

The core phenotype comprises **multisuture craniosynostosis**, **Crouzon-like midface hypoplasia** (with variable exophthalmos, relative prognathism, and a beaked nose), **delayed/failed tooth eruption with supernumerary teeth**, and **conductive hearing loss** from middle-ear anomalies. A subset of patients additionally shows a connective-tissue disorder with joint hypermobility and intervertebral disc fragility. Mechanistically, IL11RA loss of function abolishes IL-11 signaling through the shared **gp130 (IL6ST)/JAK-STAT3** pathway. Because IL-11 signaling normally *restricts* cranial suture fusion and tooth number, its loss releases that restraint, producing premature suture fusion and dental excess. This causal chain is strongly supported by human genetics, in-vitro STAT3 phosphorylation assays, structural modeling of ligand/co-receptor binding, and `Il11ra`-null mouse models that recapitulate the cranial and skeletal phenotype (with incomplete penetrance of craniosynostosis).

The disorder is **ultra-rare** (~40–50 patients reported as of recent literature), enriched in **consanguineous Turkish and Pakistani families**, with recurrent variants and low-frequency carrier alleles detectable in gnomAD. There is **no disease-specific pharmacotherapy or gene therapy**; management is **multidisciplinary and primarily surgical/supportive** (cranial vault remodeling, fronto-orbital advancement, distraction osteogenesis, plus ENT and orthodontic care). Prognosis for survival is generally favorable — patients survive to adulthood — with morbidity driven chiefly by **recurrent multifactorial intracranial hypertension**, craniofacial deformity, hearing loss, and dental complications. Prevention is limited to **genetic counseling, carrier/cascade testing, and prenatal/preimplantation diagnosis**; no primary (environmental) prevention exists because the etiology is purely monogenic.

Key Findings

1. CRSDA is an autosomal recessive syndrome caused by biallelic loss-of-function IL11RA variants (F001)

Homozygosity mapping in **three consanguineous Pakistani families** localized the syndrome to chromosome 9, and all affected children carried homozygous **IL11RA** missense mutations — `c.662C>G` (p.Pro221Arg), `c.734C>G` (p.Ser245Cys), and `c.886C>T` (p.Arg296Trp). Two North European families additionally carried a nonsense variant `c.475C>T` (p.Gln159X) and an in-frame duplication `c.916_924dup` (p.Thr306_Ser308dup) (PMID: 21741611). The original report states that *"all affected children harbored homozygous missense mutations (c.662C>G [p.Pro221Arg], c.734C>G [p.Ser245Cys], or c.886C>T [p.Arg296Trp]) in IL11RA (encoding interleukin 11 receptor, alpha) on chromosome 9p13.3."*

A second landmark study confirmed the gene in additional families, identifying a splice mutation `c.479+6T>G` *"in the interleukin 11 receptor alpha gene (IL11RA) on chromosome 9p21"* and further missense/nonsense variants, all of which *"dramatically reduced IL11-mediated STAT3 phosphorylation"* (PMID: 24498618).

Ontology mapping: Gene **HGNC:5967 (IL11RA)**; disease **MONDO:0013615 / OMIM:614188**.

2. Core phenotype: multisuture craniosynostosis, Crouzon-like midface hypoplasia, dental anomalies, and conductive hearing loss (F002)

The syndrome *"featuring craniosynostosis, maxillary hypoplasia, delayed tooth eruption, and supernumerary teeth"* (PMID: 21741611) presents with a characteristic craniofacial gestalt described as *"multiple suture synostosis, midface hypoplasia, variable degree of exophthalmos, relative prognathism, a beaked nose, and conductive hearing loss"* (PMID: 24498618). Middle-ear anomalies underlie the conductive hearing loss (PMID: 30811827). Importantly, *"a subset of patients had an associated connective tissue disorder with joint hypermobility and intervertebral discs fragility"* (PMID: 29926465).

Suggested HPO terms:

Phenotype	HPO term	Onset	Frequency
Craniosynostosis (multisuture)	HP:0001363	Congenital	Core/near-universal
Midface/maxillary hypoplasia	HP:0011800 / HP:0000327	Congenital–childhood	Frequent
Delayed tooth eruption	HP:0000684	Childhood	Frequent
Supernumerary teeth	HP:0011069	Childhood	Frequent
Conductive hearing impairment	HP:0000405	Childhood	Frequent
Proptosis / exophthalmos	HP:0000520	Childhood	Variable
Relative prognathism	HP:0000303	Childhood	Variable
Joint hypermobility	HP:0001382	Variable	Subset

3. Mechanism: IL11RA LOF abrogates IL11→gp130→JAK/STAT3 signaling, releasing restraint on suture fusion and tooth number (F003)

Functional assays demonstrated that *"the p.Arg296Trp mutation rendered the receptor unable to mediate the IL11 signal, indicating that the mutation causes loss of IL11RA function"* (PMID: 21741611). The same study concluded that *"IL11 signaling is essential for the normal*

development of craniofacial bones and teeth and that its function is to restrict suture fusion and tooth number." Across all tested variants, investigators *"demonstrated dramatically reduced IL11-mediated STAT3 phosphorylation for all mutations"* (PMID: 24498618).

Structural modeling of newer variants (p.Pro116Leu, p.Glu126Gly, p.Gly231Val, p.Leu236Pro) showed they disrupt hydrogen-bond networks, *"significantly reducing ligand-binding affinity to both interleukin-11 (IL-11) and gp130,"* alongside truncating variants p.Pro218Argfs*140 and p.Trp132Ter (PMID: 40353334).

Suggested GO/CL terms: GO:0038165 (interleukin-11-mediated signaling pathway); GO:0007259 (JAK-STAT cascade / receptor signaling via STAT); GO:0060349 (bone morphogenesis); GO:0001503 (ossification); CL:0000062 (osteoblast); CL:0000092 (osteoclast); CL:0000055 (suture mesenchyme / cranial neural-crest-derived osteoprogenitors).

4. Pathway corroboration: gp130 (IL6ST)/SOCS3 axis and an independent GP130 craniosynostosis variant (F006)

IL-11 is an IL-6 family cytokine that signals through a receptor complex incorporating **gp130**. A review of the IL-6 cytokine family notes these cytokines have *"specific functions in craniofacial development, longitudinal bone growth, and the maintenance of trabecular and cortical bone structure, and have been implicated in musculoskeletal pathologies such as craniosynostosis,"* regulated by *"their negative feedback pathways, particularly suppressor of cytokine signaling 3 (SOCS3)"* (PMID: 34332274). Independently, a variant in the **GP130 receptor subunit (IL6ST)** that selectively impairs IL11 signaling was also linked to craniosynostosis (PMID: 32566365), providing orthogonal genetic validation of the IL11–gp130 axis as the critical node. IL-11 biology and emerging therapeutics are reviewed in PMID: 31161823.

5. Variant spectrum: biallelic missense, nonsense, frameshift, splice-site, and in-frame duplication, clustering near functional motifs (F013)

The IL11RA pathogenic allele catalog spans all major variant classes, each producing loss of function:

Variant class	Examples	Source
Missense	p.Pro221Arg, p.Ser245Cys, p.Arg296Trp, p.Pro200Thr, p.Arg237Pro, p.Arg261Cys, p.Pro116Leu, p.Glu126Gly, p.Gly231Val, p.Leu236Pro	P 21741611 P 24498618 P 30811827 P 40353334
Nonsense	p.Gln159X, p.Tyr232, *p.Arg292*, p.Trp132Ter	P 21741611 P 24498618 P 40353334
Frameshift	p.Pro218Argfs*140	P 40353334
Splice-site	c.479+6T>G (aberrant transcripts)	P 24498618
In-frame duplication	c.916_924dup (p.Thr306_Ser308dup)	P 21741611 P 30811827

Several variants cluster in functionally critical regions: *"these two mutation sites cluster together near a WSXWS motif that likely plays a significant role in regulating IL11RA protein function"* (PMID: 30811827). Compound-heterozygous and homozygous configurations both occur: *"a family with pansynostosis with compound heterozygous missense mutations, p.Pro200Thr and p.Arg237Pro; and two further Turkish families with Crouzon-like syndrome carrying the homozygous nonsense mutations p.Tyr232* and p.Arg292*"* (PMID: 24498618). Per ACMG/AMP, LOF is an established mechanism for IL11RA, supporting pathogenic classification of truncating and functionally validated missense alleles.

6. Mouse and model-organism evidence recapitulates and extends the phenotype (F004, F009)

`Il11ra`-null mice show *"disturbed cranial growth and suture activity … in which reduced size and remodeling of limb bones has been previously described"* (PMID: 21741611). Direct comparison of IL11 vs IL11RA1 knockouts established that *"loss of function (LOF) in IL11RA infers IL11 signaling as important for fertility, fibrosis, inflammation and incompletely penetrant craniosynostosis"* (PMID: 34239012) — documenting both **incomplete penetrance** of craniosynostosis and the **pleiotropy** of the pathway. A classic reproductive phenotype shows *"female mice with a null mutation of the interleukin-11 receptor alpha chain are infertile because of defective decidualization"* (PMID: 9500603); human data corroborate IL-11/IL-11Rα dysregulation in endometrium of infertile women with endometriosis (PMID: 16310857).

7. Rarity and population genetics (F005)

"A total of 41 similar patients have been reported with IL11RA variants" as of 2019 (PMID: 30811827), rising to roughly 40–50 with subsequent reports. Carrier alleles are detectable at low frequency: in *"Non-Finnish Europeans … an allele frequency for p.Thr306_Ser308dup of 0.014% and p.Arg261Cys of 0.008%."* Ascertainment is enriched in consanguineous populations — three consanguineous Pakistani families (PMID: 21741611) and Turkish families, including a 12-member family from Antakya and 5 mutation-positive families among a 79-patient craniosynostosis cohort (PMID: 24498618).

Section-by-Section Characterization

1. Disease Information

CRSDA (MONDO:0013615; OMIM:614188) is a rare autosomal recessive syndromic craniosynostosis defined by craniosynostosis plus dental anomalies and Crouzon-like facial features. Synonyms: **Kreiborg–Pakistani syndrome**, IL11RA-related Crouzon-like autosomal recessive craniosynostosis. Information is derived from **aggregated disease-level resources** (OMIM, case series) built on individual patient reports in consanguineous families, not large EHR cohorts.

2. Etiology

Purely genetic/monogenic: biallelic LOF variants in IL11RA (F001, F014). There is **no established environmental, infectious, lifestyle, or teratogenic contribution** — *"all affected children harbored homozygous … mutations"* (PMID: 21741611). **Consanguinity** is the principal risk-amplifying factor. No protective variants or gene–environment interactions are described. Modifier contribution is suggested by incomplete penetrance of craniosynostosis in mice and intrafamilial phenotypic heterogeneity in humans, but specific human modifier genes are not established.

3. Phenotypes

See Finding 2 and HPO table above. Onset is **congenital** (cranial sutures) to **childhood** (dental). Severity is **variable** even within families; course is **non-episodic and lifelong/progressive** with respect to intracranial pressure and craniofacial growth. Quality-of-life impact arises from craniofacial deformity, functional visual/airway consequences of midface hypoplasia, hearing loss affecting speech/learning, and dental/orthodontic burden; no disease-specific QoL instrument data are available.

4. Genetic/Molecular Information

Causal gene **IL11RA (HGNC:5967)**, gene OMIM 600939, chr 9p13.3, UniProt Q14626. Variant spectrum per Finding 5. Functional consequence: **loss of function** (reduced STAT3 phosphorylation; reduced IL-11/gp130 binding). Inheritance biallelic (homozygous or compound heterozygous). No recurrent large chromosomal abnormality, aneuploidy, or established epigenetic lesion is implicated. gp130/IL6ST provides a second locus for a phenotypically overlapping craniosynostosis via impaired IL11 signaling (F006).

5. Environmental Information

Not applicable — no environmental, lifestyle, toxic, or infectious contributors are reported (F014). The disorder is defined solely by biallelic IL11RA LOF.

6. Mechanism / Pathophysiology — Ordered Causal Chain

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1. Biallelic LOF variant in IL11RA (missense/nonsense/frameshift/splice/in-frame dup)
     -> results in a defective or absent IL-11 receptor alpha subunit
2. Defective IL11RA -> abolishes assembly of the functional IL-11 / IL11RA / gp130 (IL6ST) compl
     -> reduces ligand-binding affinity to IL-11 and gp130 [demonstrated: structural + binding a
3. Failed receptor signaling -> abrogates JAK-mediated STAT3 phosphorylation
     -> "dramatically reduced IL11-mediated STAT3 phosphorylation" [demonstrated in HEK293T/COS7
4. Loss of STAT3 output -> removes the normal IL-11 restraint on osteogenic differentiation at s
     -> BRANCH A (bone): premature/excess suture ossification -> multisuture craniosynostosis
                         -> restricted calvarial/midface growth -> midface hypoplasia, exophthal
                            prognathism, beaked nose; middle-ear involvement -> conductive heari
     -> BRANCH B (teeth): loss of restraint on tooth number/eruption -> delayed eruption + supern
     -> BRANCH C (variable): connective-tissue effects -> joint hypermobility, disc fragility [i
5. Fixed cranial deformity + ongoing brain growth -> recurrent, multifactorial intracranial hype
     -> clinical morbidity requiring staged surgical intervention
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Steps 1–3 are **demonstrated** in vitro and by structural modeling; step 4 branches are supported by human phenotype–genotype correlation and `Il11ra`-null mouse recapitulation; the connective-tissue branch (C) and the precise cell-autonomous osteoblast/osteoclast balance are **inferred**. Negative feedback normally runs through **SOCS3** (F006). Upstream = IL11RA LOF; downstream = suture/tooth/ear phenotypes and ICP. Cell types: cranial suture mesenchyme, osteoblasts, osteoclasts, odontogenic epithelium/mesenchyme.

Pathway/ontology terms: KEGG JAK-STAT; Reactome "Interleukin-11 signaling"; GO:0038165, GO:0007259, GO:0001503, GO:0045669 (positive regulation of osteoblast differentiation).

7. Anatomical Structures Affected (F010)

- **Primary:** cranial sutures/calvaria (UBERON:0000211 suture / UBERON:0001684 skull), midface/maxilla (UBERON:0002397), teeth (UBERON:0001091), middle ear ossicles (UBERON:0001686).
- **Secondary/variable:** orbit (exophthalmos), axial skeleton/intervertebral discs (UBERON:0002412), joints/connective tissue; `Il11ra`-null mice also show reduced limb-bone size/remodeling.
- **Body systems:** skeletal, craniofacial, auditory, dental; reproductive in the mouse pleiotropy context. Lateralization: generally **bilateral/symmetric** for sutures, though suture-specific patterns vary.
- **Cell types (CL):** osteoblast (CL:0000062), osteoclast (CL:0000092), odontoblast (CL:0000060), chondrocyte (CL:0000138).

8. Temporal Development (F011)

Onset is **congenital** (premature suture fusion present from infancy) with dental features emerging through childhood. Course is **chronic, lifelong, progressive** (not episodic). *"Premature cranial base fusion correlates with recurrent increased ICP and repeated cranial vault remodeling procedures"* (PMID: 42402848). Critical intervention windows coincide with skull growth in infancy/childhood; survival to adulthood is documented (two adult brothers, PMID: 30811827).

9. Inheritance and Population (F005, F012)

Autosomal recessive — *"it is inherited according to an autosomal recessive mode of inheritance"* (PMID: 29926465); unaffected carrier parents carry **25% recurrence risk** per pregnancy. **Ultra-rare** (~40–50 reported). Enriched in **consanguineous Turkish/Pakistani** families; founder/recurrent alleles present. Penetrance of craniosynostosis is **incomplete** (mouse evidence); **expressivity is variable** with intrafamilial heterogeneity. Carrier alleles in gnomAD at 0.008–0.014% in Non-Finnish Europeans. No strong sex bias reported for the craniofacial phenotype.

10. Diagnostics (F007)

Diagnosis rests on recognizing **AR Crouzon-like syndromic craniosynostosis with dental anomalies** and confirming **biallelic IL11RA variants** by **whole-exome/targeted NGS with Sanger validation** — *"genetic analyses were carried out utilizing whole-exome sequencing, with subsequent validation through direct Sanger sequencing"* (PMID: 40353334). Homozygosity mapping plus targeted sequencing identified cases among a *"cohort of 79 patients with an initial clinical diagnosis of Crouzon syndrome, pansynostosis, or unclassified syndromic craniosynostosis"* (PMID: 24498618). **Differential diagnosis:** FGFR2-related Crouzon syndrome (AD) is the key mimic — *"two adult brothers diagnosed with Crouzon syndrome as children, in which the clinical diagnosis of CRSDA was made on reevaluation"* (PMID: 30811827); AR inheritance, dental anomalies, hearing loss, and negative FGFR testing distinguish CRSDA. Imaging (CT/3D-CT) defines suture fusion; audiometry documents conductive hearing loss. In a Turkish syndromic CS cohort, IL11RA was the gene specifically for *"craniosynostosis with dental anomaly"* (PMID: 42084887).

11. Outcome / Prognosis (F008, F014)

Survival is **generally favorable** (adulthood documented). Morbidity is driven by **recurrent multifactorial intracranial hypertension** — *"intracranial hypertension in syndromic craniosynostosis is multifactorial. Abnormal cerebrospinal fluid circulation, venous hypertension, upper airway obstruction, and premature cranial base fusion appear to contribute substantially to disease progression"* (PMID: 42402848) — plus craniofacial deformity, hearing loss, and dental complications. Crouzon-like phenotypes generally showed favorable surgical outcomes (PMID: 42084887). Repeated procedures may be required.

12. Treatment (F008)

No disease-specific pharmacotherapy or gene therapy exists. Management is **multidisciplinary, primarily surgical and supportive**, following general syndromic craniosynostosis principles: - **Cranial vault remodeling / fronto-orbital advancement** (NCIT clinical-intervention: Cranial Vault Remodeling; Fronto-Orbital Advancement). - **Distraction osteogenesis**, including **posterior vault distraction osteogenesis (PVDO)** — *"intracranial volume increased by 36.3% at device removal and 27.8% at 1-year follow-up"* (PMID: 41931363). - **Endoscope-assisted suturectomy with orthotic therapy** and **spring-assisted cranioplasty** as minimally invasive options (PMID: 42686138; PMID: 42853819). - **Midface advancement**, **ENT/audiology** (hearing amplification), and **orthodontic/oral-surgical** management of dental anomalies.

Management is notably heterogeneous across centers (PMID: 41627210). No pharmacogenomic, targeted, immunotherapy, or RNA-based therapy applies.

13. Prevention (F012)

Limited to **genetic counseling, carrier/cascade testing, and prenatal/preimplantation genetic diagnosis**; **no primary prevention** (etiology is monogenic). AR inheritance yields 25% recurrence risk; detectable gnomAD carrier alleles enable targeted screening in at-risk families, especially where **consanguinity** raises risk. Secondary/tertiary prevention = early surgical intervention and ICP surveillance to avert complications.

14. Other Species / Natural Disease (F004, F009)

Primary model species: **mouse (NCBI Taxon:10090)**, ortholog **Il11ra1**. `Il11ra`-null mice reproduce disturbed cranial growth/suture activity and reduced limb-bone size/remodeling, plus **female infertility** from defective decidualization (PMID: 9500603; PMID: 34239012). No

naturally occurring companion-animal/wildlife disease or zoonotic relevance is reported. IL11RA is a highly conserved gp130-family receptor, supporting evolutionary conservation of the mechanism.

15. Model Organisms (F004, F009)

Model	Type	Key phenotype	Limitation
`Il11ra`-null mouse	Mammalian knockout	Disturbed cranial growth/suture activity; reduced limb-bone remodeling; **incompletely penetrant craniosynostosis**; female infertility	Incomplete penetrance; does not fully capture human dental/midface gestalt
`Il11` knockout mouse	Mammalian knockout	Fibro-inflammation (lung), fertility defects, craniosynostosis	Ligand- vs receptor-level differences
In-vitro HEK293T/ COS7	Cellular	Reduced IL11-mediated STAT3 phosphorylation (functional readout)	Not developmental context

Resources: MGI, IMPC. Applications: suture biology, IL11–gp130–STAT3 signaling, bone remodeling, and the pleiotropic roles in fibrosis and reproduction.

Mechanistic Model / Interpretation

CRSDA is best understood as a **loss-of-cytokine-restraint disorder**. In normal craniofacial development, IL-11 binds IL11RA, which partners with gp130 (IL6ST) to activate JAK–STAT3; this signal **restricts** osteogenic suture fusion and tooth number, held in check by SOCS3 negative feedback. Biallelic IL11RA LOF collapses this signal at its first obligatory step, so the developmental "brake" is released — sutures fuse prematurely (craniosynostosis), the midface fails to grow forward (Crouzon-like hypoplasia), and odontogenesis over-produces/mis-times teeth. The independent gp130/IL6ST craniosynostosis variant that selectively impairs IL11 signaling confirms that the **node**, not merely the gene, is what matters.

```
IL-11 --> IL11RA --> gp130(IL6ST) --> JAK --> STAT3 --> restrains suture fusion / tooth number
                                ^                                |
                                +-------- SOCS3 (-) <----------+
   [IL11RA LOF breaks this circuit -> disinhibited ossification + dental excess]
```

The same pathway's pleiotropy (fertility via decidualization, fibro-inflammation) explains why `Il11ra`/`Il11` knockouts show phenotypes beyond the skull, and raises the clinically relevant caveat that **systemic IL-11 pathway modulation is not a simple therapeutic lever** for this developmental disorder.

Evidence Base

PMID	Role	Contribution
21741611	Landmark (human + mouse)	Identified IL11RA as causal; defined phenotype; established LOF and developmental restraint role
24498618	Confirmatory (human)	Additional families/variants; STAT3 phosphorylation defect; Crouzon differential
40353334	Recent (human + structural)	Novel variants; impaired IL-11/gp130 binding via H-bond disruption; WES diagnostics
30811827	Clinical/population	Case count (~41); gnomAD carrier frequencies; WSXWS clustering; misdiagnosis as Crouzon
29926465	Clinical series	AR inheritance; connective-tissue subset
34239012	Model organism	Incomplete penetrance; pleiotropy (fertility, fibrosis, inflammation)
9500603	Model organism	Reproductive pleiotropy (decidualization)
34332274	Review	IL-6 family/gp130 in bone; SOCS3 feedback
32566365	Corroborating gene	GP130/IL6ST variant -> craniosynostosis via impaired IL11 signaling
31161823	Review	IL-11 biology and emerging therapeutics
42402848	Surgical/clinical	Multifactorial ICP pathophysiology in syndromic CS
41931363	Surgical	PVDO intracranial-volume outcomes
42686138, 42853819, 41627210	Surgical	Distraction/spring/minimally invasive options; care heterogeneity
42084887	Cohort	IL11RA as the gene for "craniosynostosis with dental anomaly"
16310857	Human corroboration	IL-11/IL-11Rα dysregulation in infertile endometrium

Limitations and Knowledge Gaps

- **Small N:** ~40–50 patients limit precise penetrance, expressivity, sex-ratio, and QoL estimates; no formal prevalence/incidence figures exist.
- **Genotype–phenotype correlation is weak**, with documented intrafamilial heterogeneity; modifier genes in humans are unidentified.
- **Connective-tissue branch** mechanism is inferred, not mechanistically dissected.
- **No disease-specific natural-history study, registry, or validated QoL/outcome instrument.**
- **No targeted therapy;** whether IL-11 pathway augmentation could help is untested and complicated by pleiotropy.
- Some supporting surgical literature derives from general syndromic CS cohorts, not IL11RA-specific series.

Proposed Follow-up Experiments / Actions

1. **Establish an international CRSDA registry** linking genotype, suture pattern, dental findings, hearing, ICP history, and surgical outcomes.
2. **Conditional/suture-specific `Il11ra` mouse models** and iPSC-derived cranial neural crest/osteoblast organoids to dissect the osteoblast–osteoclast balance and the tooth-number phenotype.
3. **Structural/biochemical deep-mutational scanning** of IL11RA mapped onto the IL-11/IL11RA/gp130 complex to functionally classify VUS and resolve the WSXWS-motif contribution.
4. **Define the connective-tissue subset** — ECM/collagen profiling in patients with joint hypermobility and disc fragility.
5. **Prospective comparison of surgical strategies** (PVDO vs cranial vault remodeling vs minimally invasive) specifically in IL11RA-confirmed patients, with ICP and neurocognitive endpoints.
6. **Carrier-screening pilots** in high-consanguinity populations using the known recurrent alleles.